Sarthak Thakur | AI Engineer & Python Developer

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Motivated Computer Science student with a strong foundation in Python and Data Science. Demonstrated ability to clearly communicate complex technical concepts (Transformers, Neural Networks) and build functional AI tools. Passionate about applying mathematical foundations to solve real-world problems through scalable code.

Education

Shoolini University **Solan, HP**
Bachelor of Technology in Computer Science, CGPA: 8.7/10.0 *2023 – 2027 (Expected)*
Overall GPA: 8.87

Technical Skills

Languages: Python (Intermediate), SQL, HTML, CSS, JavaScript (Basic) , Rubby, Rust, GO
Data Science: NumPy, Pandas, Matplotlib, Scikit-learn
AI & NLP: HuggingFace Transformers, PyTorch, NLTK
Web Frameworks: Streamlit
Tools: Git, GitHub, VS Code, Jupyter Notebooks , Docker, AWS

Projects

Generative AI Project
GPT-2 Fine-Tuning (Football Dataset) *2025*
- Fine-tuned GPT-2 on a custom **2,000+ line football dataset**, utilizing the HuggingFace Transformers library.
- Improved training loss steadily across **4 epochs**, generating coherent domain-specific text outputs.
- Handled data preprocessing and tokenization pipelines to optimize the input data for the model.

NLP Application
Resume Analyzer Web App *2025*
- Built a **Streamlit** web app to analyze resumes using **TF-IDF & Cosine Similarity**.
- Tested the algorithm on **15+ sample resumes** to accurately compute keyword match scores against job descriptions.
- Designed a clear interface to visualize missing skills and relevance percentages for users.

Machine Learning
Movie Recommendation System *2025*
- Developed a content-based movie recommender using the **TMDB 5k dataset** (4,800+ movies).
- Implemented vectorization logic to generate **top-5 movie recommendations** with near-instant results. ○ Visualized dataset correlations using Matplotlib to understand genre clustering and user preferences.

Deep Learning Research

GPT-2 Architecture Implementation *2025* *1/2*

○ Pretrained a miniature GPT-2 model on a **10MB text corpus** to understand transformer internals.
○ Implemented core mechanisms like **Self-Attention** and **Positional Encodings** from scratch in PyTorch. ○ Documented pretraining dynamics and attention patterns in a technical report.

Self-Development

Self-Directed

DSA & Python Practice *2023 – Present* ○ Solved **50+ Data Structures and Algorithms problems** on platforms like LeetCode/GeeksForGeeks.

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○ Focused on optimizing solution efficiency for Arrays, Strings, and Linked Lists.
Maintaining a GitHub repository of Python utility scripts to practice modular coding and version control.

Relevant Coursework

○ Analysis of Algorithms
○ Data Structures
○ Probability & Statistics
○ Database Management Systems